MENGPANG XING

SOFTWARE ENGINEERING | SITE RELIABILITY | CLOUD INFRASTRUCTURE

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EDUCATION

Bachelor of Arts in Computer Science and Data Science | Northwestern University | Evanston, IL | ***Expected Jun 2027***

TECHNICAL SKILLS & KNOWLEDGE

Programming Language: C, C++, Python, Typescript/Javascript, Java, Go, R, Rust, SQL
Cloud & Infrastructure: GCP (GKE, GCE, BigQuery, Cloud Monitoring), AWS (ECS, EKS, S3), Kubernetes, Docker
DevOps & CI/CD: Argo Workflows, GitOps, Jenkins, CI/CD
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs
Frontend Development: React, TypeScript
Databases & Data: PostgreSQL, BigQuery, Pandas, NumPy
AI & Machine Learning: LLM Agents, Prompt Engineering, OpenAI API, PyTorch, TensorFlow
Reliability & Observability: Observability, Alerting & Monitoring, Incident Triage, Toil Reduction, Cloud Cost Optimization (FinOps)

CORE COMPETENCIES

Software Engineering | Data Structures | Incident Triage | LLM Agents | Site Reliability | Full-Stack Development | Cloud Infrastructure | Distributed Systems | Algorithm Design | REST API Integration | CI/CD Pipelines | Cloud Observability | Deployment Automation | Database Systems | Data Engineering | Cloud Cost Optimization | Application Programming | Prompt Design | Machine Learning | Data Quality

PROFESSIONAL EXPERIENCE

Software Engineer Intern | CME Group | Chicago, IL | Jun 2026 – Present

- Engineer an LLM-powered ticket-intake guardrail agent that validates required context, enriches submissions, and blocks malformed partner requests before entering the SRE queue, reducing manual intake work and on-call engineer time while improving request quality.
- Design a BigQuery-backed CI/CD and Deployment Health dashboard providing on-call engineers immediate root-cause isolation across GCE, network, DNS, Identity (IAM), and logging-provider issues, reducing mean time to detect (MTTD) for infrastructure deployment failures.
- Standardize failure signatures and quota/stockout triggers across log pipelines to eliminate high-volume false positives, strengthen schema handling, and isolate ***3,701*** actionable failures for more reliable live alerting.
- Automate cloud cost controls through a multithreaded Python audit tool comparing historical GCP metrics with live API state, ***identifying 78*** unused compute reservations and enabling alerts for resources idle seven or more days.
- Configure a Gemini Enterprise top-level orchestrator agent to route complex infrastructure logs through specialized data-retrieval, synthesis, and QA/compliance sub-agents, enabling layered incident analysis and daily identification of deployment failures and unused reservations.
- Publish historical peak metrics to BigQuery through the Unified Data Analytics Platform, defining YAML schemas and deployment runbooks and promoting changes across four environments, enabling consistent production access to historical infrastructure metrics.

Software Engineer Intern | Cars Commerce (Cars.com) | Chicago, IL | Mar 2025 – Aug 2025

- Built a production customer-data platform using React, FastAPI, and PostgreSQL, centralizing analytics for ***1,000+ automotive dealers***.
- Containerized and deployed on AWS ECS with Jenkins CI/CD, sustaining ***200 QPS at 50 ms*** latency while reducing release time by ***70%***.
- Migrated workloads from EKS to ECS, eliminating Kubernetes cluster-maintenance overhead while preserving application availability and enabling a more streamlined deployment environment with reduced operational complexity.
- Implemented ***16 Metaplane*** data-quality monitors across ***50+ critical tables***, reducing mean time to detect data issues by ***40%*** and safeguarding critical pricing and lead-generation pipelines through improved data reliability and visibility.

Full-Stack Developer Intern | Y STEM and Chess Inc | Boise, ID (Remote) | Jun 2024 – Sep 2024

- Developed RESTful APIs with Node.js and Express for authentication, lesson tracking, and chess game-state functionality, supporting ***500+ students*** and scalable platform operations as enrollment grew ***3x year-over-year***.
- Optimized PostgreSQL schema and queries to reduce response times by ***30%***, strengthening backend performance as usage expanded.
- Crafted responsive, accessible React and TypeScript interfaces in collaboration with the design team, enhancing the overall learning experience, improving lesson interactivity, and reducing student drop-off by ***25%***.

PROJECT PORTFOLIO

RubricXpert | AI-Powered Essay Evaluation System | Python | Flask | React | OpenAI API | Longformer | MiniLM | PyTorch | Mar 2025

- Created a parallelized AI essay-evaluation system integrating GPT-4o, Longformer, and MiniLM, leveraging ***10 concurrent*** workers to accelerate multi-model inference for rubric-aligned grading, theme extraction, and meta-analysis.
- Enhanced GPU-based inference and structured JSON outputs to deliver real-time, criterion-level feedback across **50+ rubric** criteria per essay, enabling efficient downstream analytics and dashboard workflows.